THE UNIVERSITY OF MELBOURNE

IT Capstone Project

COMP 30022 2025

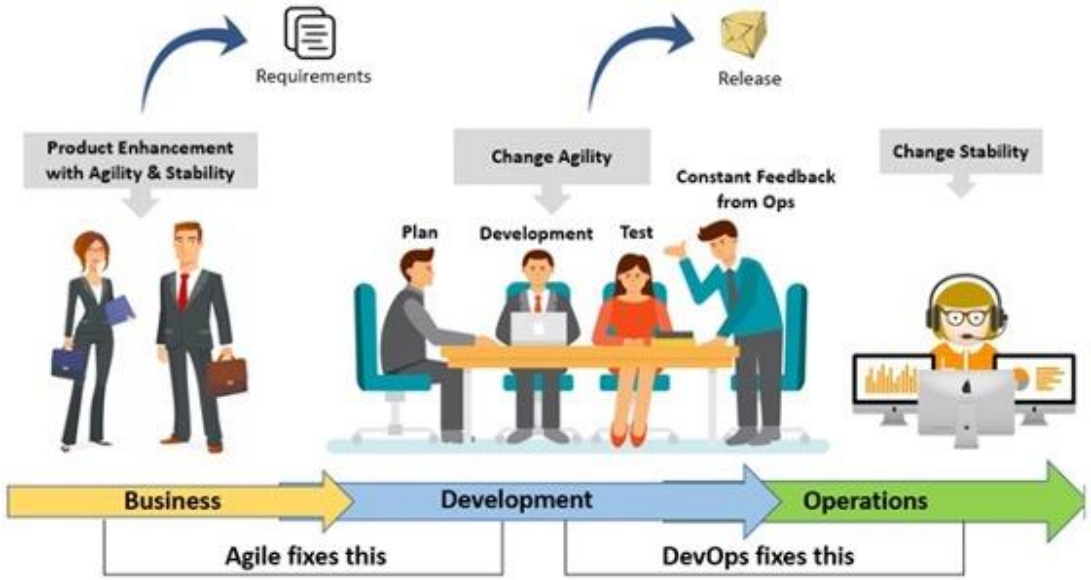
Week 8 – Deployment

Overview of lecture

- DevOps
- CI/CD
- Demo

Before DevOps

- Two teams core to delivering functional software:
  - ***Developer teams:*** develop, build, and test features.
  - ***Operations teams****:* manage the change management, security, deployment, monitoring, and feedback.
- Teams worked in isolation.
- Led to a lot of problems integrating code and managing the release.



DevOps

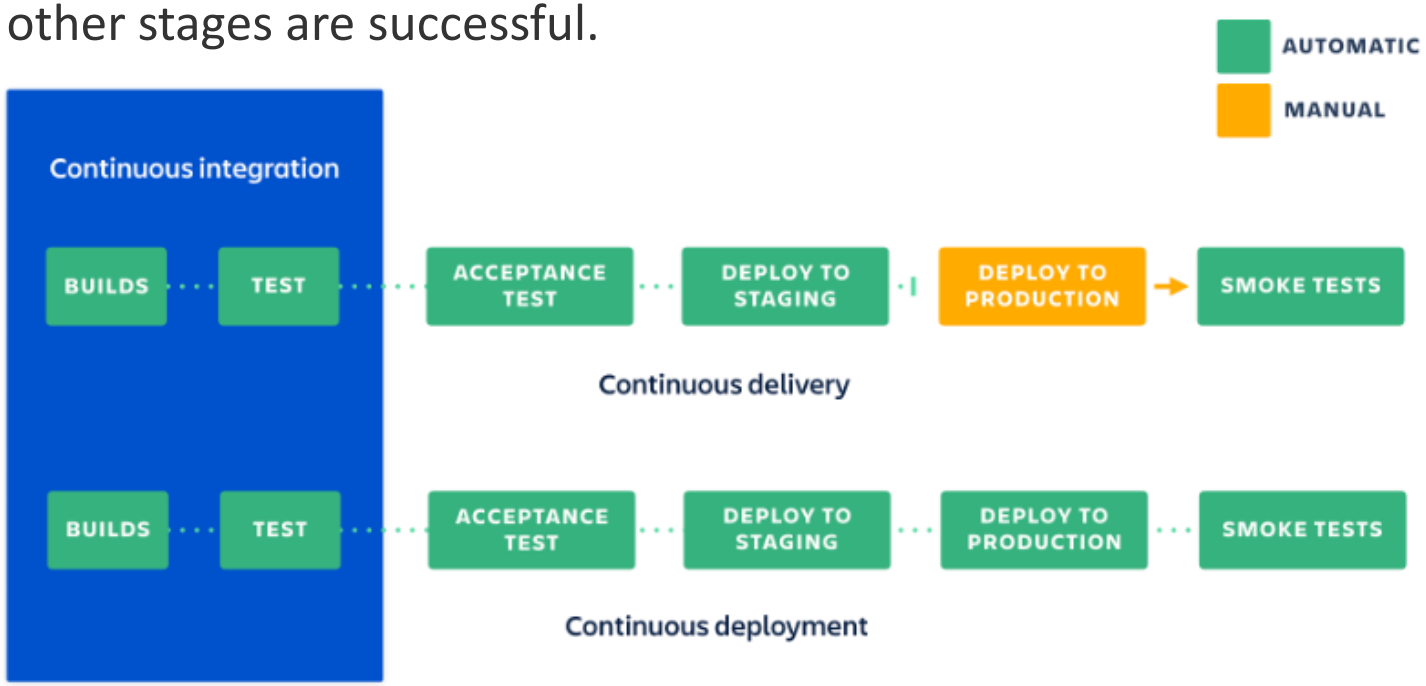
- Aims to break down these silos by enabling developer and operations teams to work together.
- Increases collaboration and allows teams to manage releases by using automated workflows.

Key Concepts

- ***Automation**:* using technology to perform a task in a reproducible way, such that feedback is provided on the process itself and minimal human intervention is required.
- ***DevOps pipeline**:* set of automated processes. Includes continuous integration, continuous delivery/deployment, etc.

CI/CD

- Continuous Integration: Integrating your code changes back to your main branch. Changes are checked by automatically building, testing code.
- Continuous Delivery: Extension of CI. Automates the release process so that you can deploy the application at any time.
- Continuous Deployment: Extension of continuous delivery. Automate deployment to production environment if all other stages are successful.

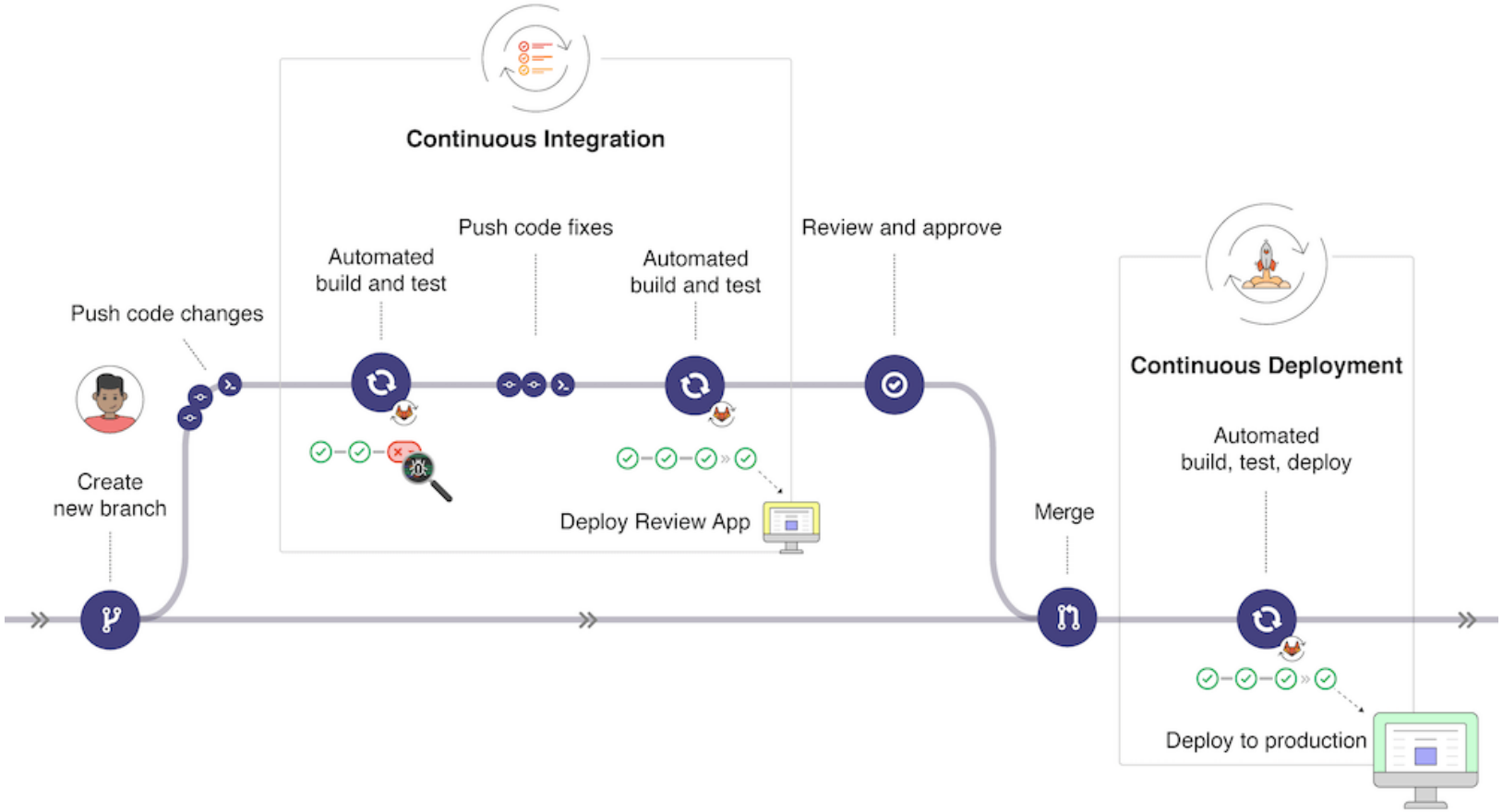


Why CI / CD?

1. **Fast**: the time taken for releases are significantly reduced through automating the workflow.
2. **Simple**: the process of integrating code becomes much simpler
3. **Fewer errors**: less intervention from developers is required which reduces the chance for human error
4. **Isolated failures**: immediate feedback provided as to which step in the pipeline has failed

DevOps Pipeline

- A set of steps executed when a condition is met (e.g. a pull request is made)
- Need to ensure checks are performed before code is deployed
- Typical workflow:



DevOps in COMP30022

- We expect you to consider DevOps and deployment options in your project
- Implementation will vary depending on your product / client needs
- For web apps, we would ideally expect an automated deployment pipeline to be set up so that stable releases from master can be deployed to a production server without the need for manual intervention.
- GitHub Actions is a good choice (although you may use a different service if you prefer)

Deployment

- Set of steps required to make your application accessible to the public
- Three main models:

SaaS	PaaS	IaaS
Software as a service.	Platform as a service.	Infrastructure as a service.
The vendor manages everything.	Vendor manages architecture, but you develop the application and manage data.	You manage the infrastructure, data, and application.

- Examples:
  - SaaS: Dropbox, Salesforce, Cisco WebEx, GoToMeeting
  - PaaS: AWS Elastic Beanstalk, Microsoft Azure, Heroku, Google App Engine
  - IaaS: DigitalOcean, Amazon Web Services (AWS), Cisco Metapod, Microsoft Azure, Google Compute Engine (GCE)
- PaaS is likely easiest for this subject

Demo